

MS CHEMSITRY RESEARCH SYNOPSIS

**Sequestration of CO₂ gas by HKUST-1@ZrP/functionalized-Silica
MOF Composite Material**



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**Sequestration of CO₂ gas by HKUST-1@ZrP/functionalized-Silica MOF
Composite Material**

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MS in (CHEMISTRY)**

A Research Proposal / Synopsis submitted to the University of Education, Lahore
in partial fulfillment of the requirements of Higher education commission for Master of
Philosophy in Chemistry.

**UNIVERSITY OF EDUCATION LAHORE, ATTOCK CAMPUS
2025-27**

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Chapter 1

INTRODUCTION

The rapid increase in the atmospheric greenhouse gas (GHG) concentration, especially carbon dioxide, has become an environmental challenge of the 21st century [1]. The continuous growth of the global population and industrialization has led to substantial increase in global energy consumption and the fossil fuel remaining the dominant source, supplying more than 85% of the world energy demand despite their significant contribution to atmospheric CO₂ emission [2]. Despite the temporary decline in energy consumption during COVID-19 pandemic, the international energy agency (IEA) project that global energy demand will continue to grow steadily in coming decades with an estimate of 1.3% annual growth until 2040 [3].

The main indicators of climate change are melting of glaciers, rise in sea levels, and acid rains [4]. These rising temperature and extreme weather threaten agriculture, reducing food production and crop yield [5]. The climate change is also causing allergies by changing the concentration of pollens, allergenic potential, accelerating the spread of allergenic plants, increasing aeroallergen levels and the risk of respiratory allergies [6]. Climate change alters the nutrients cycle by changing the temperature, precipitation, forest composition, vegetations and hydrological cycles [7]. Severe drought weaken the forest productivity and heighten vulnerability to insects, diseases and wildfire while reducing the biodiversity and economics resources [8],[9],[10],[11].

To control climate changes and greenhouse gases emission, The Paris agreement adopted by 195 countries played an important role in global climate policies after Kyoto protocol and the conference of Copenhagen 2009 failure [12]. This agreement sets a temperature goal of holding global warming below 2°C above pre-industrial level and makes efforts to limit it to 1.5°C to minimize climate impacts [13],[14],[15]. But the global temperature are still expected to rise by 2.6–3.1°C by 2100 [16]. The long term success of Paris agreement depends on mutual trust participating countries and the government take decisive actions that guide and regulate the energy markets and reduce emissions [17],[18].

Carbon capture, utilization and storage (CCUS) is essential to achieving net zero emissions. Capturing methods include pre-combustion, post-combustion and oxy-fuel combustion [19]. Low concentration capture is done by Direct Air Capture (DAC). The materials used for the carbon captured are Liquid adsorbents and Solid adsorbents [20].

Activated Carbon, Zeolites, Metal oxides, Mesoporous silica and Hydrotalcites are the major adsorbent classes [21]. Carbon based adsorbents are widely studied for carbon capture due to their abundance, tunable properties and cost effectiveness. Activated carbon, carbon nanotubes, graphene and graphene oxides, biochar and hydrochar,

carbon fiber and aerogels are the types of carbon based materials. Carbon based materials involved some challenges for their use [22].

Zeolites materials are crystalline porous solids with tunable structures. Due to their high surface area, pore uniformity and chemical stability, they are considered promising adsorbent [23]. Zeolite adsorb CO₂ via ion dipole interaction and the performance is influenced by cation type, pore size and Silica/Alumina ratio [24]. Zeolites show strong preference for CO₂ over N₂ and O₂, making them suitable for post combustion and direct air capture [25]. Water competes with CO₂ for adsorption sites and reduced the efficiency of zeolites [26].

Metal oxides absorb CO₂ by turning it into stable carbonates which can later be broken down with heat to release the CO₂ and reuse the materials [27]. Metal oxides are broadly categorized in alkaline earth oxides, alkali metal silicates and zirconates, transition metal oxides and layered double oxides [28]. Metal oxides are frequently impregnated onto carbon support to combine physisorption and chemisorption mechanism [29]. Metal oxides sorbents suffer from structural degradation, sluggish absorption and deactivations, limiting their widespread deployment in industry [30][31].

Mesoporous silica is an excellent candidate for CO₂ capture due to their high stability under heat and chemicals, high surface area and adjustable porous structure [32]. These materials are effective mainly because they featured organized pore structure, large BET surface area and can be prepared in many forms like powders, rods, sphere or discs [33]. Mesoporous silica contain numerous micro and mesopores and a high density of reactive Si-OH groups on the pore walls, offering sites for CO₂ adsorptions and enabling further chemical modification [34]. Pure Silica capture CO₂ through weak van der Waals forces with low capacity and poor selectivity. Amine functionalized silica combined physisorption and chemisorption to significantly improve the carbon uptake and selectivity [35][36]. The use of amine-functionalized mesoporous silica in industry is restricted because amine can leach out, degrade under heat and making it in bulk is still a problem [37].

Hydrotalcites also known as layered double hydroxide are promising anionic clays for CO₂ capture at intermediate temperature, which makes them useful for applications involving hot flue gas and syngas [38]. Hydrotalcites demonstrate excellent long term durability when regenerated with steam, consistently retaining their CO₂ capture capacity across multiple cycles. However, Hydrotalcites show reduced CO₂ uptake when they undergo many cycles of adsorption with CO₂ or H₂O, each followed by nitrogen flushing [39][40].

Conventional adsorbents suffer from interrelated limitations spanning cost, moisture sensitivity, high regeneration energy and cyclic reproducibility that hinder pilot scale deployment [41][42].

Metal-organic Frameworks are (MOFs) are porous crystalline materials constructed by linking metals ion or cluster with organic ligands through coordination bond [43]. The field emerged in the late 1990s, pioneered by Omar Yaghi [44]. MOFs boast among the highest pore sites ever measured in synthetic materials, with surface area frequently ranging from 5000-700 m²/g and the pores can occupy upto 90 percent of the volume of the materials

[45][46]. Metal-organic frameworks (MOF) demonstrate thermal stability typically between 250 C and 500C, attributed to their robust covalent bonding network [47]. Metal-organic frameworks (MOFs) surpass traditional porous adsorbents, achieving CO₂ uptake levels above 2870 mg/g, compared with only 310 mg/g for zeolites and 345 mg/g for ultramicroporous graphitic carbon [48].

STATEMENT OF THE PROBLEM

Global warming driven by excess CO₂ emissions is one of the most critical challenges of our time. Most of these emissions come from burning fossil fuels in power plants and industries. Today, factories use liquid chemical solutions (amine) to trap CO₂ before it reaches the atmosphere. However, this method has big drawbacks. It consumes huge amount of energy to desorb CO₂ or reuse the liquid. The chemicals corrode the metal equipment and the liquid breakdown into toxic waste overtime.

Solid materials called Metal-organic frameworks (MOFs) offers much cleaner and low energy alternatives. Because MOFs have tiny pores and large surface areas. They can act like a sponge to trap CO₂. But before we use MOFs in real factories, we have to fix few problems related to it.

- i. Factory or Chimney smoke contain a lot of moisture. Most simple MOFs absorb water instead of CO₂. Which ruins their structure or blocks them from trapping CO₂.
- ii. If we modify MOFs with strong chemicals to trap CO₂ better, they hold onto the gas too tightly. This means we end up spending too much energy heating them up to release CO₂ or regenerating the MOFs.
- iii. Factor and Chimney smoke is not pure CO₂. It is mostly mixed with Nitrogen and other available gases. Because many MOFs lack the ability to separate these gases, they trap the N₂ alongside the Carbon. This means the materials pores fill up with the wrong gases, drastically lowering its actual capacity to catch CO₂ and making the whole process inefficient.

The Main Problem to Solve

Right now, there is a big gap in creating hybrid MOFs materials combining Zirconium Phosphate (ZrP) as a supporting nanoplatform for the deposition of layer by layer HKUST-1 upto 20 layers with the addition of functionalize Silica with Para-aminobenzoic Acid (PABA) that can trap CO₂ efficiently in moist air, release it using low energy and remain strong enough for industrial use.

Objectives of the Study

In the proposed project, main objectives are to

- i. Develop strong hybrid materials by combining Zirconium Phosphate (ZrP) as a supporting nanoplatform for the deposition of layer by layer HKUST-1 up to 20 layers with the addition of functionalize Silica with Para-aminobenzoic Acid (PABA) .

- II. Study the physical and chemical properties of the prepared materials using common lab techniques (like XRD, BET, FTIR, TGA and SEM) to confirm their structure and pore size.
- III. Test carbon capture performance by measuring how much CO₂ the material can hold in dry condition.
- IV. Find the best way to reuse the materials by testing how easily the CO₂ released under mild heating, ensuring the material retains its performance.

Significance of the Study

In this work, HKUST-1, zirconium phosphate (ZrP), and functionalized silica hybrid composites will be synthesized. HKUST-1 possesses an exceptionally high specific surface area, open metal coordination sites, and excellent gas uptake capabilities. The thermal stability, chemical durability, and moisture resistance of the framework are significantly enhanced by incorporating 20 layered zirconium phosphate (ZrP) into the matrix. Functionalized silica (Functionalized by Para-aminobenzoic Acid PABA) provides superior mechanical robustness, a well-defined porous network, and tunable surface active sites while being cost effective and chemically stable. Therefore, hybrid composites of HKUST1, 20 layered zirconium phosphate (ZrP), and functionalized silica thus synthesized will combine all these characteristic properties, making them far more stable, durable, and highly efficient for carbon dioxide capture in Chimneys or factories.

Chapter 2

LITERATURE REVIEW

The Metal-organic Framework (MOF), constructed through secondary molecular building blocks called the reticular synthesis have predetermined structure, composition and properties. These materials exhibit outstanding gas sorption capacity and vast surface area, exceeding zeolite and carbon, while their internal pore geometries can be systematically tuned in a controlled manner and can be chemically modified to optimize uptake [49,50]. Due to this high surface area and tunable pore size, Metal-organic Frameworks (MOFs) are widely studied for gas adsorption, separation and catalysis [51]. Among various MOFs, $\text{Cu}_3(\text{BTC})_2$ commonly known as a HKUST-1 is one of the most frequently used material [52]. It contains open Cu^{2+} paddlewheel nodes that provide strong binding sites for polar gas molecules resulting in high adsorption capacity [53 54].

However, the main practical issue with HKUST-1 is its poor stability in the presence of moisture and it shows great affinity towards water [55]. When HKUST-1 is exposed to humidity, the water molecules readily attack the copper-carboxylate (Cu-O) coordination bond through competitive coordination. This hydrolytic attack breaks the framework structure and leading to the loss of crystallinity and complete pore collapse [56,57]. While adding hydrophobic external coatings has been tested to repel moisture [58,59]. Thick protective layers usually block the internal pores and lower the overall gas uptake [60].

To improve framework stability without blocking the pores, supporting MOFs on stable inorganic matrices has become a common strategy [61]. Layered α -zirconium phosphate

($\alpha\text{-Zr}(\text{HPO}_4)_2 \cdot \text{H}_2\text{O}$, or $\alpha\text{-ZrP}$) is particularly suitable for this purpose [62]. It possesses high thermal stability and strong mechanical durability and good resistance across different chemical environment [63].

The basal plane of $\alpha\text{-ZrP}$ sheets contain active P-OH groups that allow easy surface modification and coordination [64]. When used as a solid substrate, ZrP nanosheets help disperse MOF crystals uniformly, prevent particular agglomeration and provide a rigid mechanical backbone that reduces structural stress during gas adsorption desorption cycles [65].

Direct growth of MOFs on untreated inorganic support often results in poor interfacial contact and phase separation [66,67]. Therefore, modifying the ZrP surface with organic functional groups (such as amine, carboxyl, or silane group) is necessary [68,69]. These attached functional groups act as chemical anchors that coordinate

directly with Cu^{2+} ions, guiding the uniform nucleation and growth of HKUST-1 [70]. By using a stepwise layer by layer growth method, thin and uniform layers of HKUST-1 can be deposited over the functionalized ZrP sheets [71,72]. This approach provides strong interfacial bonding between the support and the MOF, protecting the copper nodes against moisture attack while maintaining accessible pore channels for target molecules [73].

Chapter 3

Research Methodology

Required Chemicals/Apparatus

Following materials and apparatuses will be used during Experimental performance:

- Sample vial
- Beaker
- Crucible plate
- Aluminum sheet
- Conical Flask
- Funnel
- Spatula
- China Dish
- Glass Stirrer
- Watch glass
- Round bottom flask
- Three neck Flask
- Silicon flexible tubes
- Stand
- Valve
- Ballon
- Thermometer
- CO₂ gas Cylinder
- N₂ gas cylinder
- Weighing balance
- Ultrasonic bath
- Furnace
- Vacuum drier
- HKUST-1@ZrP
- Silica
- Para-aminobenzoic acid (PABA)
- Ethanol
- Methanol

Procedure

Step 1: Functionalization of Silica

Weigh the silica in crucible and put it in the furnace for 08 hours at 480C. Weigh a known amount of Para-aminobenzoic acid (PABA) and introduce it to 250ml distill water

in a round bottom flask and stir for 5 minutes to get a homogeneous solution. Add the calcinated silica to it under constant stirring. Reflux this disperse solution for 30 minutes under control thermal conditions. After the reflux, cover the solution with aluminum foil and leave it undisturbed for 24h at room temperature. Dissolve the formed crystals by the addition of methanol and place the solution in the oven to form the dry powder. A functionalized Silica-amine powder will be obtained.

Step 2: Synthesis of HKUST-1@ZrP/f-silica Composite

Activate the functionalized silica by adding a known amount in the 5ml ethanol and Sonicate it for 5 minutes. Introduce a stoichiometric amount of pre-synthesized HKUST1@ZrP in activated functionalized silica and further sonicate it for 30 minutes to get desired HKUST-1@ZrP/f-Silica MOF composite.

Step 3: Gas adsorption performance (CO₂ uptake) study

Pass the CO₂ gas through this MOF composite and study the gas uptake on different parameters, i.e temperature, pressure. Find the maximum uptake and confirm the maximum uptake equilibrium by repeating the adsorption until the uptake become constant or drop.

Step 4: Evaluation of Thermal Regeneration Conditions

Identify the minimum temperature for the desorption of MOFs composite and prevent the thermal framework degradation. The estimated temperature for the desorption HKUST1@ZrP/f-Silica is between 60-90 Celsius.

Characterization

HKUST-1@ZrP/f-Silica MOFs composite will be synthesized and then characterized by using the following instrumental techniques:

- I. FTIR spectroscopy
- II. XRD Technique
- III. SEM Microscopy technique
- IV. BET
- V. TGA Analysis technique

Research Plan

Time schedule:

Chapter 4

References

- 1) Mahajan, S., & Lahtinen, M. (2022). Recent Progress in Metal-organic Frameworks (MOFs) for CO₂ Capture At Different Pressures. *Journal of Environmental Chemical Engineering*. <https://doi.org/10.1016/j.jece.2022.108930>
- 2) Ghanbari, T., Abnisa, F., & Wan Daud, W. M. A. (2020). A review on production of metal organic frameworks (MOF) for CO₂ adsorption. *Science of The Total Environment*, 707, 135090. <https://doi.org/10.1016/j.scitotenv.2019.135090>
- 3) Mahajan, S., & Lahtinen, M. (2022). Recent Progress in Metal-organic Frameworks (MOFs) for CO₂ Capture At Different Pressures. *Journal of Environmental Chemical Engineering*. <https://doi.org/10.1016/j.jece.2022.108930>
- 4) Kumar, V. (2024). Global Warming and Climate Change. *Journal of Research in Social Science and Humanities*. <https://doi.org/10.56397/jrssh.2024.03.03>
- 5) Bilgili, M., Tumse, S., & Nar, S. (2024). Comprehensive Overview on the Present State and Evolution of Global Warming, Climate Change, Greenhouse Gasses and Renewable Energy. *Arabian Journal for Science and Engineering*, 49, 14503 - 14531. <https://doi.org/10.1007/s13369-024-09390-y>
- 6) D'Amato, G., & Akdis, C. (2020). Global warming, climate change, air pollution and allergies. *Allergy*, 75. <https://doi.org/10.22541/au.159526736.69654469>
- 7) Bandh, S. A., Shafi, S., Peerzada, M., Rehman, T., Bashir, S., Wani, S. A., & Dar, R. (2021). Multidimensional analysis of global climate change: a review. *Environmental Science and Pollution Research*, 28, 24872 - 24888. <https://doi.org/10.1007/s11356-021-13139-7>
- 8) P.E. Kauppi, J.H. Ausubel, J. Fang, A.S. Mather, R.A. Sedjo, & P.E. Waggoner, Returning forests analyzed with the forest identity, *Proc. Natl. Acad. Sci. U.S.A.* 103 (46) 17574-17579, <https://doi.org/10.1073/pnas.0608343103>
- 9) Keeley, Jon & Fotheringham, Cj. (2001). Historic Fire Regime in Southern California Shrublands. *Conservation Biology*. 15. 1536 - 1548. [10.1046/j.15231739.2001.00097.x](https://doi.org/10.1046/j.15231739.2001.00097.x)

- 10) Keenan, R.J. Climate change impacts and adaptation in forest management: a review. *Annals of Forest Science* 72, 145–167 (2015).
<https://doi.org/10.1007/s13595-014-0446-5>
- 11) Fleming, R.A., Candau, J.N. & McAlpine, R.S. Landscape-Scale Analysis of Interactions between Insect Defoliation and Forest Fire in Central Canada. *Climatic Change* 55, 251–272 (2002). <https://doi.org/10.1023/A:1020299422491>
- 12) FALKNER, R. (2016). The Paris Agreement and the new logic of international climate politics. *International Affairs*, 92(5), 1107-1125.
<https://doi.org/10.1111/14682346.12708>
- 13) Rogelj, J., Elzen, M., Höhne, N., Fransen, T., Fekete, H., Winkler, H., Schaeffer, R., Sha, F., Riahi, K., & Meinshausen, M. (2016). Paris Agreement climate proposals need a boost to keep warming well below 2 °C. *Nature*, 534, 631-639.
<https://doi.org/10.1038/nature18307>
- 14) Höhne, N., Kuramochi, T., Warnecke, C., Röser, F., Fekete, H., Hagemann, M., Day, T., Tewari, R., Kurdziel, M., Sterl, S., & Gonzales, S. (2017). The Paris Agreement: resolving the inconsistency between global goals and national contributions. *Climate Policy*, 17, 16 - 32.
<https://doi.org/10.1080/14693062.2016.1218320>
- 15) Mor, S., Aneja, R., Madan, S., & Ghimire, M. (2023). Kyoto Protocol and Paris Agreement: Transition from Bindings to Pledges – A Review. *Millennial Asia*, 15, 690 - 711. <https://doi.org/10.1177/09763996221141546>
- 16) Rogelj, J., Elzen, M., Höhne, N., Fransen, T., Fekete, H., Winkler, H., Schaeffer, R., Sha, F., Riahi, K., & Meinshausen, M. (2016). Paris Agreement climate proposals need a boost to keep warming well below 2 °C. *Nature*, 534, 631-639. <https://doi.org/10.1038/nature18307> [17] Mor, S., Aneja, R., Madan, S., & Ghimire, M. (2023). Kyoto Protocol and Paris Agreement: Transition from Bindings to Pledges – A Review. *Millennial Asia*, 15, 690 - 711.
<https://doi.org/10.1177/09763996221141546>
- 17) Cléménçon, R. (2016). The Two Sides of the Paris Climate Agreement. *The Journal of Environment & Development*, 25, 24 - 3.
<https://doi.org/10.1177/1070496516631362>
- 18) Dziejarski, B., Serafin, J., Andersson, K., & Krzyżyńska, R. (2023). CO2 Capture Materials: A Review of Current Trends and Future Challenges. *Materials Today Sustainability*. <https://doi.org/10.1016/j.mtsust.2023.100483>
- 19) Rana, A., & Andino, J. M. (2025). A Review of Materials for Carbon Dioxide Capture. *Catalysts*. <https://doi.org/10.3390/catal15030273>

- 20) Petrovic, B., Gorbounov, M., & Soltani, S. (2020). Influence of surface modification on selective CO₂ adsorption: A technical review on mechanisms and methods. *Microporous and Mesoporous Materials*, 110751. <https://doi.org/10.1016/j.micromeso.2020.110751>
- 21) Sharma, A., Jindal, J., Mittal, A., Kumari, K., Maken, S., & Kumar, N. (2021). Carbon materials as CO₂ adsorbents: a review. *Environmental Chemistry Letters*, 19, 875 - 910. <https://doi.org/10.1007/s10311-020-01153-z>
- 22) Donglong Fu, Mark E. Davis; Carbon dioxide capture with zeotype materials. *Chem. Soc. Rev.* 2022; 51 (22): 9340–9370. <https://doi.org/10.1039/d2cs00508e>
- 23) Liao, G., He, Y., Wang, H., Fang, B., Tsubaki, N., & Li, C. (2023). Carbon neutrality enabled by structure-tailored zeolite-based nanomaterials. *Device*. <https://doi.org/10.1016/j.device.2023.100173>
- 24) Fu, D., Park, Y., & Davis, M. E. (2022). Confinement effects facilitate lowconcentration carbon dioxide capture with zeolites. *Proceedings of the National Academy of Sciences of the United States of America*, 119. <https://doi.org/10.1073/pnas.2211544119>
- 25) Murge, P., Dinda, S., & Roy, S. (2019). Zeolite-Based Sorbent for CO₂ Capture: Preparation and Performance Evaluation.. *Langmuir : the ACS journal of surfaces and colloids*. <https://doi.org/10.1021/acs.langmuir.9b02259>
- 26) Dziejarski, B., Serafin, J., Andersson, K., & Krzyżyńska, R. (2023). CO₂ Capture Materials: A Review of Current Trends and Future Challenges. *Materials Today Sustainability*. <https://doi.org/10.1016/j.mtsust.2023.100483>
- 27) Chang, R., Wu, X.-G., Cheung, O., & Liu, W. (2022). Synthetic solid oxide sorbents for CO₂ capture: state-of-the art and future perspectives.
- 28) Nowrouzi, M., Younesi, H., & Bahramifar, N. (2018). Superior CO₂ capture performance on biomass-derived carbon/metal oxides nanocomposites from Persian ironwood by H₃PO₄ activation. *Fuel*. <https://doi.org/10.1016/j.fuel.2018.03.035>
- 29) Rohde, R. C., Carsch, K. M., Dods, M. N., Jiang, H. Z. H., McIsaac, A., Klein, R., Kwon, H., Karstens, S. L., Wang, Y., Huang, A. J., Taylor, J. W., Yabuuchi, Y., Tkachenko, N., Meihaus, K. R., Furukawa, H., Yahne, D. R., Engler, K. E., Bustillo, K. C., Minor, A. M., . . . Long, J. R. (2024). High-temperature carbon dioxide capture in a porous material with terminal zinc hydride sites.. *Science*, 386 6723, 814-819 . <https://doi.org/10.1126/science.adk5697>
- 30) Mutch, G., Shulda, S., Mccue, A., Menart, M. J., Ciobanu, C., Ngo, C., Anderson, J., Richards, R., & Vega-Maza, D. (2018). Carbon Capture by Metal Oxides: Unleashing the Potential of the (111) Facet.. *Journal of the American Chemical Society*, 140 13,

4736-4742 . <https://doi.org/10.1021/jacs.8b01845>

- 31) Bae, J. Y., Jang, S., Cho, J., & Kang, M. (2025). Amine-Functionalized Mesoporous Silica for Efficient CO₂ Capture: Stability, Performance, and Industrial Feasibility. International Journal of Molecular Sciences, 26. <https://doi.org/10.3390/ijms26094313>
- 32) Reddy, S. B. M., Ponnammam, D., Sadasivuni, K. K., Kumar, B., & Abdullah, A. (2021). Carbon dioxide adsorption based on porous materials. RSC Advances, 11, 12658 - 12681. <https://doi.org/10.1039/d0ra10902a>
- 33) Tumurbaatar, O., Popova, M., Mitova, V., Shestakova, P., & Koseva, N. (2023). Engineering of Silica Mesoporous Materials for CO₂ Adsorption. Materials, 16. <https://doi.org/10.3390/ma16114179>
- 34) Bae, J. Y., Jang, S., Cho, J., & Kang, M. (2025). Amine-Functionalized Mesoporous Silica for Efficient CO₂ Capture: Stability, Performance, and Industrial Feasibility. International Journal of Molecular Sciences, 26. <https://doi.org/10.3390/ijms26094313>
- 35) Tumurbaatar, O., Popova, M., Mitova, V., Shestakova, P., & Koseva, N. (2023). Engineering of Silica Mesoporous Materials for CO₂ Adsorption. Materials, 16. <https://doi.org/10.3390/ma16114179>
- 36) Bae, J. Y., Jang, S., Cho, J., & Kang, M. (2025). Amine-Functionalized Mesoporous Silica for Efficient CO₂ Capture: Stability, Performance, and Industrial Feasibility. International Journal of Molecular Sciences, 26. <https://doi.org/10.3390/ijms26094313>
- 37) Figueiredo, A. P., Soria, M. A., Madeira, L. M., & Rocha, C. (2024). Preparation and Characterization of Materials for Low- to Intermediate-Temperature CO₂ Adsorption. Processes. <https://doi.org/10.3390/pr12112403>
- 38) Xin, K., Boon, J., Van Dijk, H., & Van Sint Annaland, M. (2024). Stability of potassiumpromoted hydrotalcites for CO₂ capture over numerous repetitive adsorption and desorption cycles. Frontiers in Chemical Engineering. <https://doi.org/10.3389/fceng.2024.1272152>
- 39) Xin, K., Boon, J., Van Dijk, H., & Van Sint Annaland, M. (2024). Stability of potassiumpromoted hydrotalcites for CO₂ capture over numerous repetitive adsorption and desorption cycles. Frontiers in Chemical Engineering. <https://doi.org/10.3389/fceng.2024.1272152>

- 40) Lai, J., Ngu, L., & Hashim, S. S. (2021). A review of CO₂ adsorbents performance for different carbon capture technology processes conditions. *Greenhouse Gases: Science and Technology*. <https://doi.org/10.1002/ghg.2112>
- 41) Azmi, N. Z. M., Buthiyappan, A., Raman, A., Patah, M., & Sufian, S. (2022). Recent Advances in Biomass Based Activated Carbon for Carbon Dioxide Capture - A Review. *Journal of Industrial and Engineering Chemistry*. <https://doi.org/10.1016/j.jiec.2022.08.021>
- 42) Pang, J., Zhang, Z., Zhang, S., Guo, X., Chen, Q., Zhang, X.-W., Zhou, H.-L., Gong, W., Shah, S. S. A., Zhong, C., Li, J.-R., Zhang, J.-P., Cui, Y., Jiang, H., & Bu, X.-H. (2024). Recent progress in metal-organic frameworks: part I—material preparation. *Science China Chemistry*, 68, 1230 - 1286. <https://doi.org/10.1007/s11426-0242457-y>
- 43) Yusuf, V. F., Malek, N., & Kailasa, S. K. (2022). Review on Metal–Organic Framework Classification, Synthetic Approaches, and Influencing Factors: Applications in Energy, Drug Delivery, and Wastewater Treatment. *ACS Omega*, 7, 44507 - 44531. <https://doi.org/10.1021/acsomega.2c05310>
- 44) Wang, A., Walden, M., Ettlinger, R., Kiessling, F., Gassensmith, J., Lammers, T., Wuttke, S., & Peña, Q. (2023). Biomedical Metal-Organic Framework Materials: Perspectives and Challenges. *Advanced functional materials*, 34. <https://doi.org/10.1002/adfm.202308589>
- 45) Wiśniewska, P., Haponiuk, J., Saeb, M., Rabiee, N., & Bencherif, S. (2023). Mitigating Metal-Organic Framework (MOF) Toxicity for Biomedical Applications. *Chemical engineering journal (Lausanne, Switzerland : 1996)*, 471. <https://doi.org/10.1016/j.cej.2023.144400>
- 46) Wiśniewska, P., Haponiuk, J., Saeb, M., Rabiee, N., & Bencherif, S. (2023). Mitigating Metal-Organic Framework (MOF) Toxicity for Biomedical Applications. *Chemical engineering journal (Lausanne, Switzerland : 1996)*, 471. <https://doi.org/10.1016/j.cej.2023.144400>
- 47) Wang, A., Walden, M., Ettlinger, R., Kiessling, F., Gassensmith, J., Lammers, T., Wuttke, S., & Peña, Q. (2023). Biomedical Metal-Organic Framework Materials: Perspectives and Challenges. *Advanced functional materials*, 34. <https://doi.org/10.1002/adfm.202308589>
- 48) Yaghi, O., O'Keeffe, M., Ockwig, N. et al. Reticular synthesis and the design of new materials. *Nature* 423, 705–714 (2003). <https://doi.org/10.1038/nature01650>
- 49) Hiroyasu Furukawa *et al.*, The Chemistry and Applications of Metal-Organic Frameworks. *Science* **341**, 1230444 (2013). DOI: [10.1126/science.1230444](https://doi.org/10.1126/science.1230444)
- 50) Li, H., Eddaoudi, M., O'Keeffe, M., & Yaghi, O. M. (1999). Design and synthesis of an exceptionally stable and highly porous metal-organic framework. *Nature*, 402(6759), 276–279. <https://doi.org/10.1038/46248>

- 51) Chui, S. S.-Y., Lo, S. M.-F., Charmant, J. P., Orpen, A. G., & Williams, I. D. (1999). A chemically functionalizable nanoporous material $[\text{Cu}_3(\text{TMA})_2(\text{H}_2\text{O})_3]_n$. *Science*, 283(5405), 1148–1150. <https://doi.org/10.1126/science.283.5405.1148>
- 52) Andrew R. Millward, Omar M. Yaghi; Metal–Organic Frameworks with Exceptionally High Capacity for Storage of Carbon Dioxide at Room Temperature. *Journal American Chemical Society* 28 December 2005; 127 (51): 17998–17999. <https://doi.org/10.1021/ja0570032>
- 53) Prestipino, C., Regli, L., Vitillo, J. G., Bonino, F., Damin, A., Lamberti, C., Zecchina, A., Solari, P. L., Kongshaug, K. O., & Bordiga, S. (2006). Local structure of framework Cu(II) in HKUST-1 metallorganic framework: Spectroscopic characterization upon activation and interaction with adsorbates. *Chemistry of Materials*, 18(5), 1337–1346. <https://doi.org/10.1021/cm052191g>
- 54) Küsgens, P., Rose, M., Senkovska, I., Fröde, H., Henschel, A., Siegle, S., & Kaskel, S. (2009). Characterization of metal-organic frameworks by water vapour sorption. *Microporous and Mesoporous Materials*, 120(3), 325–330. <https://doi.org/10.1016/j.micromeso.2008.11.020>
- 55) Nicholas C. Burtch, Himanshu Jasuja, Krista S. Walton; Water Stability and Adsorption in Metal–Organic Frameworks. *Chem. Rev.* 22 October 2014; 114 (20): 10575–10612. <https://doi.org/10.1021/cr5002589>
- 56) Low, J. J., Benin, A. I., Jakubczak, P., Abrahamian, J. F., Faheem, S. A., & Willis, R. R. (2009). Virtual High Throughput Screening Confirmed Experimentally: Porous Coordination Polymer Hydration. *Journal of the American Chemical Society*, 131(43), 15834–15842. <https://doi.org/10.1021/ja9061344>
- 57) Decoste, J. B., Peterson, G. W., Jasuja, H., Glover, T. G., Huang, Y. G., & Walton, K. S. (2013). Stability and degradation mechanisms of metal-organic frameworks containing the $\text{Zr}_6\text{O}_4(\text{OH})_4$ secondary building unit. *Journal of Materials Chemistry A*, 1, 5642–5650. <https://doi.org/10.1039/c3ta10662D>
- 58) Xie, L.-H., Xu, M.-M., Liu, X.-M., Zhao, M.-J., & Li, J.-R. (2020). Hydrophobic Metal -Organic Frameworks: Assessment, Construction, and Diverse Applications. *Advanced Science* (Weinheim, Baden-Wurttemberg, Germany), 7, 1901758. <https://doi.org/10.1002/advs.201901758>
- 59) Li, H., Li, L., Lin, R. B., Zhou, W., Zhang, Z., Xiang, S., & Chen, B. (2019). Porous metal-organic frameworks for gas storage and separation: Status and

- challenges. In *EnergyChem* (Vol. 1). Elsevier B.V.
<https://doi.org/10.1016/j.enchem.2019.100006>
- 60) Zhu, Q. L., & Xu, Q. (2014). Metal-organic framework composites. In *Chemical Society Reviews* (Vol. 43, pp. 5468–5512). Royal Society of Chemistry.
<https://doi.org/10.1039/c3cs60472a>
- 61) Clearfield, A. (1988). Role of Ion Exchange in Solid-State Chemistry. *Chemical Reviews*, 88, 125–148. <https://doi.org/10.1021/cr00083a007>
- 62) Liu, J., Wan, Y., Xiao, B., et al. (2024). The preparation and performance analysis of zirconium-modified aluminum phosphate-based high-temperature (RT–1500 °C) resistant adhesive for joining alumina in extreme environment. *Journal of Advanced Ceramics*, 13, 911–932.
<https://doi.org/10.26599/jac.2024.9220906>
- 63) Bashir, A., Ahad, S., Malik, L. A., Qureashi, A., Manzoor, T., Dar, G. N., & Pandith, A. H. (2020). Revisiting the Old and Golden Inorganic Material, Zirconium Phosphate: Synthesis, Intercalation, Surface Functionalization, and Metal Ion Uptake. In *Industrial and Engineering Chemistry Research* (Vol. 59, pp. 22353–22397). American Chemical Society. <https://doi.org/10.1021/acs.iecr.0c04957>
- 64) Donnadio, A., Pica, M., Nocchetti, M., & Piermatti, O. (2024). Zirconium Phosphates and Phosphonates: Applications in Catalysis. In *Catalysts* (Vol. 14). Multidisciplinary Digital Publishing Institute (MDPI).
<https://doi.org/10.3390/catal14100733>
- 65) Shekhah, O., Wang, H., Kowarik, S., Schreiber, F., Paulus, M., Tolan, M., Sternemann, C., Evers, F., Zacher, D., Fischer, R. A., & Wöll, C. (2007). Step-by-step route for the synthesis of metal-organic frameworks. *Journal of the American Chemical Society*, 129, 15118–15119. <https://doi.org/10.1021/ja076210u>
- 66) Denny, M. S., & Cohen, S. M. (2015). In Situ Modification of Metal-Organic Frameworks in Mixed-Matrix Membranes. *Angewandte Chemie - International Edition*, 54, 9029–9032. <https://doi.org/10.1002/anie.201504077>
- 67) Jahan, M., Bao, Q., Yang, J. X., & Loh, K. P. (2010). Structure-directing role of graphene in the synthesis of metal-organic framework nanowire. *Journal of the American Chemical Society*, 132, 14487–14495.
<https://doi.org/10.1021/ja105089w>
- 68) Petit, C., Burrell, J., & Bandoz, T. J. (2011). The synthesis and characterization of copper-based metal-organic framework/graphite oxide composites. *Carbon*, 49, 563–572. <https://doi.org/10.1016/j.carbon.2010.09.059>
- 69) Kalaj, M., Bentz, K. C., Ayala, S., Palomba, J. M., Barcus, K. S., Katayama, Y., & Cohen, S. M. (2020). MOF-Polymer Hybrid Materials: From Simple Composites to

Tailored Architectures. In Chemical Reviews (Vol. 120, pp. 8267–8302).

American

Chemical Society. <https://doi.org/10.1021/acs.chemrev.9b00575>

- 70) Doherty, C. M., Buso, D., Hill, A. J., Furukawa, S., Kitagawa, S., & Falcaro, P. (2014). Using functional nano- and microparticles for the preparation of metal-organic framework composites with novel properties. Accounts of Chemical Research, 47,

396–405. <https://doi.org/10.1021/ar400130a>

- 71) Zacher, D., Shekhah, O., Wöll, C., & Fischer, R. A. (2009). Thin films of metal-organic frameworks. Chemical Society Reviews, 38, 1418–1429.

<https://doi.org/10.1039/b805038b>

- 72) Yuan, S., Feng, L., Wang, K., Pang, J., Bosch, M., Lollar, C., Sun, Y., Qin, J., Yang, X., Zhang, P., Wang, Q., Zou, L., Zhang, Y., Zhang, L., Fang, Y., Li, J., & Zhou, H. C. (2018). Stable Metal–Organic Frameworks: Design, Synthesis, and Applications. In Advanced Materials (Vol. 30). Wiley-VCH Verlag.

<https://doi.org/10.1002/adma.201704303>

